A Finite-Rate Selection Law for Annealed Gibbs Sampling

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Abstract

A sampler can generate plausible examples in the wrong proportions. We show how this happens in annealed Gibbs sampling, which updates one variable at a time while lowering the temperature. In the Curie–Weiss model, binary variables tend to share one sign, giving two modes, and a weak field makes one mode more probable. Each run commits to a mode near the critical temperature, while the field still competes with random fluctuations, and rarely switches afterwards. The samples therefore keep only a fraction of the model's log-odds between the modes. We turn the classical probability of this commitment, due to Kondepudi and Nelson, into a law for the fraction that agrees closely with an exact calculation of the sampler. The law also explains why slower cooling helps little: on a linear ramp, cooling sixteen times more slowly only doubles the fraction. Reheating runs that settled away from the favored mode gives them a second choice, and the exact calculation shows that it again keeps only a fraction. In a Hopfield model, competition among stored patterns raises this fraction; a Gaussian approximation of the competition predicts the result to within 11% across pattern counts and sizes.[1]

1. Introduction

Suppose a model of handwritten digits assigns twice as much probability to "3" as to "8". A sampler should then produce about twice as many "3"s as "8"s over many runs. It can produce recognizable examples of both digits and still get this proportion wrong, and Monte Carlo estimates from its samples are then biased. How does the sampler change these proportions?

We study annealed Gibbs sampling, which redraws one variable at a time from its conditional distribution. At high temperature the distribution is broad; at low temperature its probability concentrates in modes, regions of likely outcomes separated by unlikely ones. As annealing lowers the temperature, the barriers between modes grow and the sampler stops mixing between them: later updates mostly sharpen the mode each run selected and rarely undo the selection [9].

The sampler can therefore get the mode proportions wrong even when the model is right. To measure this error alone, we use the Curie–Weiss (CW) model, whose probabilities, and those of its Gibbs sampler, can be computed exactly. Its binary variables tend to align, giving two modes at low temperature: most variables positive, or most negative. A weak field favors the positive mode, but near the critical temperature it competes with random fluctuations as each run commits to a mode.

1. Code and details: https://github.com/ali-izhar/finite-rate-selection-law

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We connect a classical prediction of the mode selection [6, 7, 10] to this error in mode proportions. The resulting law gives the fraction of the model's log-odds between the modes that the samples keep, and shows why slower cooling helps little. Extended to a Hopfield model, whose modes are stored patterns and their negatives, the argument predicts that competition among patterns raises this fraction. Tested after reheating, which lets runs that settled away from the favored mode select again, the prediction closely matches three-pattern runs but misses by up to 11% at other pattern counts or sizes.

2. An exact model of mode selection

The model's state, or *configuration*, is a vector $x = (x_1, \dots, x_N)$ of N binary variables $x_i \in \{-1, +1\}$, called spins. The CW model gives higher probability to configurations whose spins agree:

$$\pi_{\beta,G}(x) \propto \exp\left[\frac{\beta}{2(N-1)} \sum_{i \neq j} x_i x_j + \beta G \sum_i x_i\right]. \tag{1}$$

The *inverse temperature* β sets how strongly spins prefer to agree: at $\beta = 0$ all configurations are equally likely, and increasing β, which is cooling, concentrates probability on aligned configurations. The *field* G, fixed during a run, adds a preference for one sign. At fixed β, the Gibbs sampler converges to the *equilibrium distribution* $\pi_{\beta,G}$. The *magnetization* $m = N^{-1} \sum_i x_i$ summarizes a configuration: $m > 0$ means mostly positive spins, $m < 0$ mostly negative. Without a field, as $N \to \infty$, the equilibrium m concentrates at 0 for $\beta < 1$ and at two opposite nonzero values for $\beta > 1$. These two groups of configurations are the two *modes*, and $\beta = 1$ marks the *transition*, at the critical temperature. A positive field makes the positive mode more probable.

A run starts at some $\beta < 1$. Each Gibbs update redraws one uniformly chosen spin; updates occur at random times, N per unit time on average, and this unit is an *expected sweep*. A cooling schedule raises β past the transition to a final value, by which the run has settled into one mode. Configurations with the same number k of positive spins are equally probable and change k at the same rates, which lets k alone describe the run: we track $N + 1$ probabilities, one for each $k = 0, \dots, N$, instead of 2^N, one for each configuration [8]. We compute them exactly, without sampling noise, by numerically solving the linear equations that govern their change (Appendix A).

Let q_{chain} be the probability that the run ends with $m > 0$, and q_{eq} the probability of $m > 0$ under the equilibrium distribution at the final β, both conditioned on $m \neq 0$. The fraction of the model's log-odds that the samples keep is

$$c = \frac{\operatorname{logit} q_{\text{chain}}}{\operatorname{logit} q_{\text{eq}}}, \qquad \operatorname{logit} q = \log \frac{q}{1-q}, \tag{2}$$

for $q_{\text{eq}} \neq 1/2$: $c = 1$ reproduces the equilibrium proportions, and $c = 0$ gives equal mode probabilities. If equilibrium odds are 4:1, $c = 0.2$ gives sampled odds of $4^{0.2}$:1 $\approx$ 1.32:1.

3. The selection law

Each run selects its mode in a short window near $\beta = 1$. There, a small random imbalance between positive and negative spins grows into one mode (Figure 1a); afterwards, switching between modes

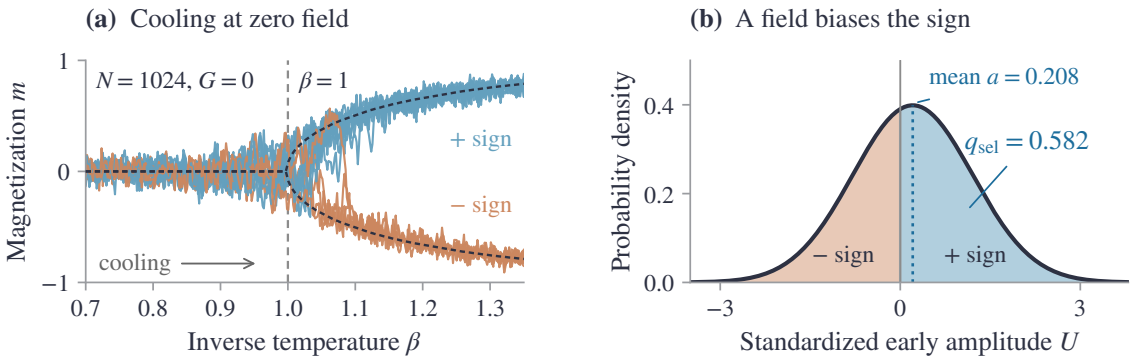


Figure 1: (a) Fourteen paths ($N = 1024$, $G = 0$) of an approximate drift-plus-noise equation with drift $-m + \tanh(\beta m)$ split into two modes, colored by final sign; the dashed curves mark the most probable equilibrium magnetization. (b) The analytical density of $U \sim \mathcal{N}(a, 1)$ for $N = 2048$, $\gamma = 0.001$ and $G = 0.0006125$.

takes far longer than settling within one [5, 8]. The final proportions are therefore decided in this window, the only time the field can bias them.

Near the transition, two effects move m; Appendix A derives both. The *drift*, the mean rate of change of m, is $-m + \tanh(\beta m + \beta G)$, approximately $(\beta - 1)m + G$ near $m = 0$ and $\beta = 1$. Below $\beta = 1$ its first term points against m and pulls an imbalance back toward zero; above $\beta = 1$ it points along m and amplifies the imbalance. The *noise* comes from the randomness of the Gibbs updates and adds a variance of about $2/N$ per expected sweep. On a linear ramp at cooling rate $\gamma > 0$, $\beta(t) = 1 + \gamma t$ crosses $\beta = 1$ at $t = 0$. The growth rate γt then itself rises with time, and the drift multiplies the imbalance by $e^{\gamma t^2/2}$, a factor that becomes large after a time T of order $\gamma^{-1/2}$: this is the window. Within it, the field adds an imbalance of order GT and the noise adds an imbalance of order $\sqrt{T/N}$ [3]. Their ratio, of order $G\sqrt{N}\,\gamma^{-1/4}$, decides the race: slower cooling widens the window and favors the field, but only as the fourth root of the cooling time.

To turn the field-to-noise ratio into the probability that a run ends in the positive mode, we divide m by its growth factor $e^{\gamma t^2/2}$. The result, the early *amplitude*, keeps the run's final sign and stops changing once the window closes. It is the field's push, the same in every run, plus the accumulated noise, a sum of many small independent Gaussian kicks; in the linear approximation it is therefore Gaussian across runs, with mean set by the field and spread by the noise. In units of its spread, it follows $U \sim \mathcal{N}(a, 1)$, where a is the field-to-noise ratio of the race with its exact prefactor. A run ends with $m > 0$ exactly when $U > 0$, which happens with probability $\Phi(a)$ (the blue area in Figure 1b), where Φ is the standard Gaussian distribution function. For the parameters of Figure 1b, $a = 0.208$: the noise is about five times stronger than the field, yet 58% of runs end in the favored mode. This is the classical result [6, 7, Appendix A]

$$q_{\text{sel}} \simeq \Phi(a), \qquad a = G\sqrt{N}(\pi/\gamma)^{1/4}, \tag{3}$$

with q_{sel} the probability of selecting $m > 0$. Without later switching, $q_{\text{chain}} = q_{\text{sel}}$.

Eq. (2) turns q_{sel} into c, giving the *selection law*. For a weak field ($|a| \ll 1$), the sampled log-odds are $\operatorname{logit} \Phi(a) \approx 4a/\sqrt{2\pi}$, proportional to a, hence to $G\sqrt{N}\,\gamma^{-1/4}$. The equilibrium log-odds at the

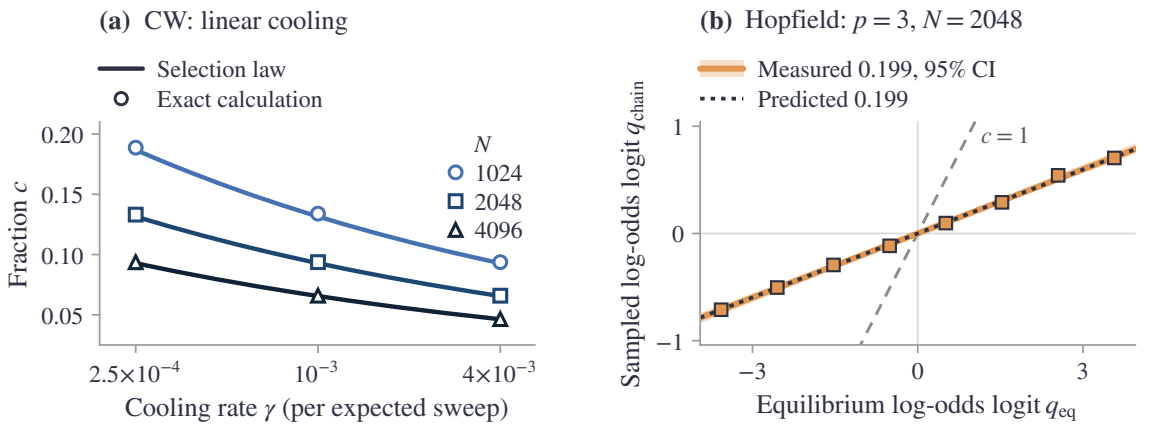


Figure 2: (a) CW ramps at fixed NG: $16\times$ slower cooling doubles c; the law is within 2%. (b) Hopfield samples keep about a fifth of the equilibrium log-odds. Each point pools twenty pattern sets; error bars are smaller than the symbols. The orange 95% band of the fitted slope c excludes reference-model uncertainty and contains the dotted prediction $A_3 c_{\mathrm{CW}}$.

final temperature grow instead as NG: reversing all N spins leaves the coupling term unchanged and reverses the field term, βGNm, of order NG in either mode. In their ratio, the field cancels, and c measures the sampler, not the field's strength (Appendix A gives the prefactor):

$$c \propto (N^2\gamma)^{-1/4}. \tag{4}$$

In this regime, cooling sixteen times more slowly doubles c; a fourfold larger N halves it.

The law rests on three assumptions (Appendix A): a slow ramp ($\gamma \ll 1$), a start well below $\beta = 1$, and little switching after selection. The noise must also stay small enough for the linear drift to hold, which $N\gamma \gg 1$ guarantees [3]. The third assumption limits how slow the cooling can be, since long enough sampling restores the equilibrium proportions, but a zero-field switch between modes takes on average about 5×10^7 expected sweeps already at $N = 2048$ and $\beta = 1.1$ (Appendix A).

Exact check. On linear ramps ($N = 1024$–4096, $\gamma = 0.00025$–0.004, NG fixed), exact c exceeds the law by only 0.23–1.81% (Figure 2a; Table 2), even at $N\gamma = 0.256$, where $N\gamma \gg 1$ fails.

4. When several modes compete

CW has two modes. To ask how competition among several modes changes the fraction c, we use the Hopfield model: with $p \ll N$ stored patterns $\xi^\mu \in \{-1, +1\}^N$ of independent equiprobable entries, its modes at low temperature are the patterns and their negatives [1]. A configuration x has probability proportional to $e^{-\beta E_G(x)}$, with *energy*

$$E_G(x) = -\frac{1}{2N}\sum_{i\neq j}\sum_{\mu=1}^{p}\xi_i^\mu\xi_j^\mu x_i x_j - G\sum_i \xi_i^1 x_i. \tag{5}$$

The field G favors pattern 1 over its negative. The overlap $m^\mu = N^{-1}\sum_i \xi_i^\mu x_i$ measures how closely a configuration resembles pattern μ: it is $+1$ at pattern μ and -1 at its negative. Each sample

is labeled $(\mu, \pm)$ by the pattern with the largest $|m^\mu|$ and the sign of that overlap. Here q_{chain} and q_{eq} are positive-sign probabilities among samples assigned to pattern 1, not pattern frequencies.

Competition raises the fraction. Each run ends in the pattern whose early overlap is largest in magnitude, and the field biases only the sign of m^1. As in Section 3, the early overlap of pattern 1 is a Gaussian shifted by the field, and at a given magnitude its sign log-odds grow linearly in that magnitude, because the Gaussian falls ever more steeply away from zero. The winner of p competing overlaps has a larger magnitude than a single overlap does, and its sign is therefore more strongly biased. To first order in the field, the bias grows by the ratio of the two average magnitudes:

$$A_p = \frac{\mathbb{E}\max_{1\le\mu\le p}|Z_\mu|}{\mathbb{E}|Z|}, \qquad c_{\text{pred}} = A_p c_{\text{CW}}, \tag{6}$$

where Z and the Z_μ are independent standard Gaussians: $|Z_\mu|$ is the size of pattern μ's early overlap in units of its spread. The numerator is the average size of the winning overlap among p patterns, the denominator that of a single overlap, as in CW, where the one pattern always wins. Here c_{CW} is the exact CW fraction for the Hopfield reheating protocol below, not for a linear ramp. Appendix C derives Eq. (6); numerical integration gives $A_1 = 1$, $A_2 = \sqrt{2}$, $A_3 = 1.662$, and $A_4 = 1.836$. The prediction assumes independent Gaussian early overlaps with a common variance, a field shifting only the mean of m^1, a final pattern decided by the largest $|m^\mu|$, and equal Hopfield and CW equilibrium sign preferences. Even for $p \ll N$, neither the independence nor this equality is guaranteed.

Reheating after a filtered start. We test the prediction where it matters for a sampler: on runs that settled away from the favored mode and are reheated to select again. Hopfield runs use *permutation sweeps*, redrawing every spin once in fresh random order (N updates, like an expected sweep), and cool through a *schedule* from $\beta = 0.6$ to 1.6, set at $N = 2048$ and reused at every size (Table 3).

Each run has three stages: a brief cooling at two sweeps per temperature, after which we remove the runs that ended in the favored mode, $(1, +)$, leaving a *filtered start*; a *reheat* of 30 sweeps at $\beta = 0.7$, where the drift pulls imbalances back toward zero; and a *final cooling* through the schedule, in which the runs select again. The same $(1, +)$ removal applies at $G < 0$; a memory of it would break the spin-flip symmetry $q_{\text{chain}}(G) + q_{\text{chain}}(-G) = 1$. The measured sums are consistent with 1 (Table 4), which detects no memory but does not show that reheated runs behave like freshly cooled ones. For this protocol, exact CW calculation gives the $c_{\text{CW}} = 0.120$ of Eq. (6) at $N = 2048$; sixteenfold longer final cooling raises it only 1.48-fold, not the twofold of Eq. (4) (Table 3).

Result. For three patterns at $N = 2048$, Eq. (6) predicts $c_{\text{pred}} = A_3 c_{\text{CW}} = 1.662 \times 0.120 = 0.199$, without any Hopfield samples. Unlike CW's, the Hopfield equilibrium log-odds are estimated from the mean overlap in separate zero-field runs, by the spin-reversal argument of Section 3 (Appendix B).

If c is a property of the sampler, sampled and equilibrium log-odds are proportional as the field varies, with slope c; the eight nonzero fields of the *field grid* approximately show this (Figure 2b). Slopes for twenty independent pattern sets, weighted by their pattern-1 samples, average to $c = 0.199$ (95% interval $[0.1908, 0.2095]$), matching the prediction. Equal pattern-set weights give 0.2101, just above the interval (Appendix B). The agreement may partly reflect cancelling errors (Section 5).

5. Where the prediction departs

The three-pattern match does not hold everywhere: across pattern counts and sizes, the prediction misses by up to 11% (Table 1; Appendix C treats $p = 2$, the largest miss, by an exact decomposition). For $p = 3$, the single-field estimate of Table 1, 0.201, agrees with the field-grid fit, 0.199.

Table 1: Measured c pools $\pm G$ against the exact CW reference (Appendix B). Predictions $A_p c_{\mathrm{CW}}$ use $c_{\mathrm{CW}} = 0.171, 0.120, 0.090$ at $N = 1024, 2048, 4096$; 95% intervals resample pattern sets. All rows use $|G| = 0.0006125$ (not rescaled with N).

p	N	predicted c	measured c
1	2048	0.120	$0.127\ [0.124, 0.131]$
2	2048	0.169	$0.188\ [0.184, 0.191]$
3	2048	0.199	$0.201\ [0.194, 0.210]$
4	2048	0.220	$0.211\ [0.191, 0.239]$
3	1024	0.284	$0.263\ [0.247, 0.283]$
3	4096	0.150	$0.160\ [0.155, 0.165]$

With one pattern there is no competition ($A_1 = 1$), and the Hopfield runs should reproduce c_{CW}; instead, separate, larger one-pattern runs give $1.063\, c_{\mathrm{CW}}$ at $N = 2048$ (Appendix B). The two samplers differ in update order (permutation versus expected sweeps) and interaction strength ($1/N$ versus $1/(N-1)$), and the prediction ignores both. Applying the same factor at $p = 3$ gives 0.212, outside the field-grid interval; the three-pattern match would then need an offsetting error of similar size.

The size dependence also departs. Measured c falls as $N^{-0.36}$ over $N = 1024$–4096, the prediction as $N^{-0.46}$ (two-point slopes, without uncertainties; Eq. (4) gives $N^{-1/2}$ for a fixed ramp), and the two cross near $N = 2048$, where the schedule was set. Scaling the prediction by the one-pattern ratio at each size does not remove this trend (Appendix B). These fixed-schedule comparisons do not measure how Hopfield sampling depends on cooling rate.

Pattern correlations are one untested explanation of the size trend: random patterns have correlations of order $N^{-1/2}$, which shift the β at which the overlaps begin to grow by as much as the *critical window*, the range of β over which fluctuations blur the transition at finite N (Appendix C).

6. Discussion

The selection probability is classical [6, 7]; the paper turns it into the fraction c of the model's log-odds that the samples keep, checks it against the exact CW chain, and extends it to competing modes. Cooling studies of Ising chains [11] measure a different error, misaligned regions and excess energy, rather than mode probabilities.

Annealed importance sampling weights annealed runs instead of reselecting them, to estimate equilibrium averages [9]; comparing it with reheating at equal cost would need valid weights for the filtered start and an assessment of estimator bias and variance. Diffusion models with an exact score reverse a noising process without equilibrating at each step [4]; extending the calculation here needs their reverse dynamics and discrete updates. In diffusion models, a variance-preserving noise schedule can miss mode asymmetry, while suitable schedules recover mode probabilities in Gaussian mixtures and CW [2].

The exact CW calculation scores any cooling schedule by its error in mode probabilities, without sampling noise; applying c to other models requires their equilibrium log-odds. Measuring Hopfield overlaps during cooling could show whether pattern correlations explain the remaining discrepancies.

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Appendix A. Selection formula and exact-chain calculations

This appendix derives the selection law of Section 3, states when it holds, and describes the exact calculation that tests it.

From spin updates to drift and noise. Section 3 describes m by a drift and a noise, and both follow from single-spin updates. For a chosen spin, let $\ell_i = \beta(N-1)^{-1}\sum_{j\neq i} x_j + h$, where $h = \beta G$. Holding the other spins fixed, Eq. (1) assigns weights e^{ℓ_i} and $e^{-\ell_i}$ to its two values, so the redrawn spin has mean $\tanh \ell_i$. For large N, the neighbors' average is approximately m. A uniformly chosen spin has old mean m, so one update changes m by $[\tanh(\beta m + h) - m]/N$ in expectation. With N updates per expected sweep, the drift is $\dot{m} = -m + \tanh(\beta m + h)$. Near $m = 0$ and in a weak field, an update flips a spin with probability about one half, and each flip changes m by $2/N$. The noise variance per expected sweep is therefore approximately $N \times \frac{1}{2} \times (2/N)^2 = 2/N$. A short interval contains many updates that change m little; we approximate the centered sum of their jumps by a Gaussian. Expanding $\tanh z = z - z^3/3 + \cdots$ near $\beta = 1$, retaining the field, linear, and cubic terms, and writing G for $h = \beta G$ (equal at $\beta = 1$) gives

$$dm = \big[(\beta - 1)m - m^3/3 + G\big]\,dt + \sqrt{2/N}\,dW_t, \tag{7}$$

where W_t is a standard Brownian motion. At $G = 0$, Eq. (7) is the swept pitchfork bifurcation analyzed by Berglund and Gentz [3]: one stable value becomes two.

Removing growth without changing the sign. On a linear ramp, $\beta - 1 = \gamma t$, the growth of m can be divided out without changing its sign. While m is small, omit the cubic term. Without field or noise, $\dot{m} = \gamma t m$ has the positive growth factor $e^{\gamma t^2/2}$. Set $Y_t = e^{-\gamma t^2/2} m_t$ and start at $t_0 < 0$. The product rule gives

$$Y_t = e^{-\gamma t_0^2/2} m_{t_0} + G\int_{t_0}^{t} e^{-\gamma u^2/2}\,du + \sqrt{2/N}\int_{t_0}^{t} e^{-\gamma u^2/2}\,dW_u.$$

The three terms are the remaining initial imbalance, the accumulated field, and the accumulated noise. With $w(u) = e^{-\gamma u^2/2}$, the noise integral is the limit of sums $\sum_j w(u_j)\sqrt{\Delta u}\,\epsilon_j$ of independent standard Gaussians ϵ_j: a Gaussian with mean zero and variance $\sum_j w(u_j)^2 \Delta u$. The field term therefore uses w, and the noise variance uses w^2.

The weights are appreciable over times of order $\gamma^{-1/2}$ around the transition. An early start suppresses the initial term; a late end lets both integrals settle. Extending their limits to the whole real line gives the amplitude Y with

$$\mu_Y = \mathbb{E}Y = G\int_{-\infty}^{\infty} e^{-\gamma u^2/2}\,du = G\sqrt{2\pi/\gamma},$$
$$V_Y = \mathrm{Var}(Y) = \frac{2}{N}\int_{-\infty}^{\infty} e^{-\gamma u^2}\,du = \frac{2}{N}\sqrt{\pi/\gamma}.$$

Hence $U = Y/\sqrt{V_Y} = a + Z$, where $Z \sim \mathcal{N}(0,1)$ and $a = \mu_Y/\sqrt{V_Y} = G\sqrt{N}(\pi/\gamma)^{1/4}$. The positive mode is selected when $U > 0$, with probability $\Pr(Z > -a) = \Phi(a)$, as in Eq. (3). These infinite limits approximate a finite cooling interval; the integrals must settle before the cubic term substantially changes the amplitude.

From a sign probability to retained log-odds. Eq. (2) then turns $\Phi(a)$ into c. For small a, $\Phi(a) = 1/2 + a/\sqrt{2\pi} + O(a^3)$. The derivative of logit q at $q = 1/2$ is 4, so logit $\Phi(a) = 4a/\sqrt{2\pi} + O(a^3)$. At the final inverse temperature β_f, a field adds $\beta_f NGm$ to each configuration's log weight. To first order in G, the positive mode's log mass changes by its average, $\beta_f NG\bar{m}_f$, and the negative mode's by the opposite amount. Here $\bar{m}_f$ is the zero-field mean magnetization in the positive mode at β_f. The equilibrium log-odds is thus approximately $2\beta_f NG\bar{m}_f$. Dividing the two leading log-odds gives

$$c \simeq \frac{\sqrt{2}\,\pi^{-1/4}}{\beta_f \bar{m}_f}(N^2\gamma)^{-1/4}.$$

Regime. With a bounded, the linear approximation and the separation from the critical window both follow from $N\gamma \gg 1$. Over the selection time $T \sim \gamma^{-1/2}$, the linear coefficient γT is of order $\sqrt{\gamma}$, so the cubic term matters once $|m| \sim \gamma^{1/4}$; the noise accumulated over T, of order $\sqrt{T/N} \sim N^{-1/2}\gamma^{-1/4}$, is a fraction $(N\gamma)^{-1/2}$ of this scale, and the field's contribution, a fraction $|G|\gamma^{-3/4} = \pi^{-1/4}a(N\gamma)^{-1/2}$ of this scale, is small too when a stays bounded. Selection must also span more than the critical window: to leading order in N, Stirling's formula and the interaction $N\beta m^2/2$ give $\log \pi(m) \simeq \text{constant} + N[(\beta-1)m^2/2 - m^4/12]$, so at $\beta = 1$ fluctuations reach $|m| \sim N^{-1/4}$ and the quadratic term matters when $|\beta - 1| \sim N^{-1/2}$, the critical window of Section 5. The selection range $\gamma T \sim \sqrt{\gamma}$ exceeds it when $N\gamma \gg 1$. These conditions allow linear selection before nonlinear growth fixes the magnitude; they do not exclude later switching.

Exact reduction and numerical evolution. To test the law, we compute the sampler's distribution exactly. Waiting times between CW updates are independent exponentials with mean $1/N$, so the count k of positive spins is a Markov chain whose rates, at a given temperature, depend only on k. Each spin updates once per expected sweep on average, so each rate is the number of eligible spins ($N - k$ negative or k positive) times their redraw probability:

$$W_k^+ = \frac{N-k}{2}\left[1 + \tanh\left(\beta\frac{2k-N+1}{N-1} + h\right)\right], \tag{8}$$

$$W_k^- = \frac{k}{2}\left[1 - \tanh\left(\beta\frac{2k-N-1}{N-1} + h\right)\right], \tag{9}$$

where $h = \beta G$; the $+1$ and -1 exclude the updated spin from the average of its neighbors. Equal probability flow between adjacent counts, $\pi_k W_k^+ = \pi_{k+1} W_{k+1}^-$ (*detailed balance*), holds for

$$\pi_k \propto \binom{N}{k} \exp\left[\frac{\beta}{2(N-1)}\left((2k-N)^2 - N\right) + h(2k-N)\right], \tag{10}$$

where the binomial factor counts configurations with the same k. The conditional reference is $q_{\text{eq}} = \sum_{k>N/2}\pi_k / \sum_{k\neq N/2}\pi_k$; the omitted zero-magnetization mass is below 10^{-36} at $\beta_f = 1.6$ for $N \geq 512$. We evolve the count probabilities $P_k(t)$,

$$\dot{P}_k = W_{k-1}^+ P_{k-1} + W_{k+1}^- P_{k+1} - (W_k^+ + W_k^-)P_k,$$

by implicit Euler steps with Richardson extrapolation. The result therefore has no sampling error.

Table 2: Linear-ramp comparison with the full selection formula. Entries give numerical c and, in parentheses, $100(c_{\mathrm{num}}/c_{\mathrm{sel}} - 1)$.

N	$\gamma = 0.004$	$\gamma = 0.001$	$\gamma = 0.00025$
1024	0.09370 (0.93%)	0.13393 (1.81%)	0.18868 (1.03%)
2048	0.06587 (0.46%)	0.09372 (0.97%)	0.13314 (1.23%)
4096	0.04645 (0.23%)	0.06589 (0.49%)	0.09361 (0.86%)

Linear ramps. The ramps of Section 3 start in the equilibrium distribution at $\beta = 0.7$ and cool to $\beta = 1.6$, with $G = 0.0006125 \times 2048/N$; a fixed NG keeps the equilibrium log-odds nearly constant across sizes (time steps: Table 5). Table 2 compares the resulting c_{num} with the full-formula prediction $c_{\mathrm{sel}} = \operatorname{logit} \Phi(a)/\operatorname{logit} q_{\mathrm{eq}}$, with a from Eq. (3). The excess over c_{sel} halves when N doubles at $\gamma = 0.004$ and 0.001, but not at 0.00025. We did not derive the correction that causes this excess.

Fixed schedule. The Hopfield prediction needs c_{CW} for the protocol of Section 4, computed the same way. Mirroring that protocol, the CW evolution starts with $\mathrm{Binomial}(N, 1/2)$ counts, runs two sweeps at each temperature of Table 3, removes $k \geq N/2$ (including $m = 0$) and renormalizes, reheats for 30 sweeps at $\beta = 0.7$, then applies Table 3. The removal keeps $k < N/2$ for both field signs, as the Hopfield runs remove $(1, +)$ for both, so the two logits in c_{CW} share the sign of G. Switching after selection is unlikely while each dwell is much shorter than the mean time for the exact zero-field chain to pass between mode centers: at $N = 2048$, about $10^{7.7}$, $10^{22.9}$, and $10^{137.8}$ expected sweeps at $\beta = 1.1$, 1.2, and 1.6, against 834 sweeps in the whole schedule. Eq. (4) describes a constant-rate ramp from a start well below $\beta = 1$, whereas this schedule dwells at ten fixed temperatures after a filtered start and reheating; we did not isolate which difference makes sixteenfold final cooling raise c less than twofold (Table 3).

Schedule design. The dwells of that schedule come from relaxation times. The relaxation time τ is the inverse *spectral gap*, the smallest positive decay rate of the chain at fixed temperature: in time τ, the slowest deviation from equilibrium shrinks by e^{-1}. The six-τ dwell of Table 3 is a relaxation prescription, not a bound on the error in final mode probabilities. To measure relaxation within a sign, the *reflected* chain (even N) keeps $k = N/2, \ldots, N$, sets $W^{-}_{N/2} = 0$, and leaves the other rates unchanged. At and below the transition, the schedule uses the full chain, because reflection would remove the slow relaxation of the balance between signs. At $\beta = 1$ and $G = 0$, the full-chain inverse gap is 42.2, 59.0, and 82.7 sweeps for $N = 1024$, 2048, and 4096; it grows as $\sqrt{N}$, so dwells redesigned for each N would too.

Appendix B. Simulation and statistical methods

This appendix describes how the Hopfield measurements of Sections 4 and 5 are made and how their uncertainty is estimated.

Sampling and labels. The Hopfield runs use Gibbs updates for the energy of Eq. (5): each update redraws x_i with probability $\Pr(x_i = +1 \mid x_{-i}) = [1 + \tanh(\beta H_i)]/2$, where x_{-i} denotes the

Table 3: **Cooling schedule.** The dwell is $\lceil 6\tau_{\mathrm{des}} \rceil$ sweeps (we chose the multiplier six), where τ_{des} is the zero-field CW inverse spectral gap of the full chain for $\beta \leq 1$ and of the reflected chain above. Counts computed at $N = 2048$ are reused at every size; Hopfield runs apply them as permutation sweeps, the CW evolution as expected sweeps. For the CW evolution at $G = 0.0006125$, halving the time step from 0.02 to 0.01 changes c_{CW} by 2.8×10^{-11}; over the eight fields, c_{CW} ranges from 0.119498 to 0.119808. Multiplying only the final cooling's counts by 0.25 and 4 gives $c = 0.098978$ and 0.146625 (ratio 1.4814).

β	.60	.80	.90	.95	1.00	1.05	1.10	1.20	1.40	1.60
sweeps	15	30	59	107	355	145	65	31	16	11

Table 4: Complete field grid, pooled over twenty pattern sets per row. Of $40{,}000$ initial runs per row, removing $(1, +)$ leaves 32,467–34,178 to reheat (33,331 at $G = 0$); n finish assigned to pattern 1 (ties go to the first pattern index: 101 initial, no final ties), and n_+ of these have positive sign. At $G = 0$, the Wilson interval is $[0.4863, 0.5048]$. The last column is the 95% interval of $q_{\mathrm{chain}}(G) + q_{\mathrm{chain}}(-G)$, with the two Wilson half-widths added in quadrature on each side; it does not resample shared pattern sets or adjust for four comparisons. After the reheat, the CW full-chain inverse gap ($N = 2048$, $\beta = 0.7$, $G = 0$) gives $e^{-30/\tau_{\mathrm{full}}} \approx 1.2 \times 10^{-4}$; this does not bound conditional Hopfield memory.

$10^4 G$	n	n_+	q_{chain}	reference q_{eq}	paired sum
6.125	11085	7417	.6691	.9726	$[0.98620, 1.01058]$
4.375	11131	7038	.6323	.9275	$[0.99624, 1.02136]$
2.625	11110	6355	.5720	.8220	$[0.98614, 1.01209]$
0.875	11033	5783	.5242	.6248	$[0.98221, 1.00852]$
0	11220	5560	.4955	.5	
−0.875	11080	5221	.4712	.3752	
−2.625	11229	4796	.4271	.1780	
−4.375	11617	4374	.3765	.0725	
−6.125	11801	3886	.3293	.0274	

other spins and $H_i = N^{-1} \sum_\mu \xi_i^\mu \sum_j \xi_j^\mu x_j - (p/N) x_i + G\xi_i^1$; the term $-(p/N)x_i$ removes self-interaction. All runs begin with independent unbiased spins, and the field is present during the initial cooling, reheating, and final cooling. Only final samples labeled pattern 1 with nonzero sign (*retained* samples) enter the estimate. Every nonzero-field point's sample interval lies strictly between $1/2$ and its reference interval; all eight nonzero fields of Table 4 enter the fit, and none was excluded. Table 5 lists seeds and run counts.

Equilibrium reference. For each pattern set s, 256 runs start at $+\xi^1$ and run at $\beta_f = 1.6$, $G = 0$. After 60 equilibration sweeps, overlaps are recorded for 200 sweeps; all measurements were labeled $(1, +)$. Their mean $\bar{m}_s^1$ gives the weak-field reference $\operatorname{logit} q_{\mathrm{eq},s}(G) \approx 2\beta_f N G \bar{m}_s^1$. The standard error (SE) divides the standard deviation of the pattern set's 256×200 overlaps by $\sqrt{256 \times 200/\tau_{\mathrm{ref}}}$, with $\tau_{\mathrm{ref}} = 1.6946$ the reflected CW relaxation time at $N = 2048$, $\beta = 1.6$, $G = 0$. τ_{ref} is an

approximate correlation correction, fixed before the runs, not a measured Hopfield correlation time. The standard error of the 256 independent run means, which needs no correlation time, is 0.85–1.05 times the reported SE for each pattern set. The log-odds envelope adds the statistical term $1.96\,\mathrm{SE}$ and a 1% systematic allowance for the reference, a design choice rather than a measured bound (a common 1% increase in the overlaps reduces c by about 1%):

$$\Delta L = 2\beta_f N|G|\big[1.96\,\mathrm{SE}(\bar{m}^1) + 0.01\bar{m}^1\big].$$

The pooled references in Table 4 and Figure 2b weight pattern set s by $\omega_s = n_s(G)/\sum_r n_r(G)$, with $n_s(G)$ its retained samples at field G: $\bar{m}^1 = \sum_s \omega_s \bar{m}^1_s$ and $\mathrm{SE}(\bar{m}^1)^2 = \sum_s \omega_s^2 \mathrm{SE}(\bar{m}^1_s)^2$. In Figure 2b, this envelope (both terms) gives the horizontal error bars, and pooled 95% Wilson intervals give the vertical ones. The bootstrap excludes the allowance and error in the Hopfield equilibrium approximation. The weak-field formula itself is accurate: for CW at $N = 2048$ and $\beta = 1.6$, it differs from the exact q_{eq} by at most 2.108×10^{-8} over the eight fields.

Slope estimate. With this reference, each pattern set gives a slope, and the slopes are averaged. For field j and pattern set s, let $x_{js} = \operatorname{logit} q_{\mathrm{eq},js}$, $y_{js} = \operatorname{logit} \widehat{q}_{js}$ with $\widehat{q}_{js} = n_{+,js}/n_{js}$, and $[L_{js}, U_{js}]$ the 95% Wilson interval of $\widehat{q}_{js}$ [12]. Each pattern set's slope weights its measurements by the inverse squared logit half-width of this interval:

$$w_{js} = \left[\frac{\operatorname{logit} U_{js} - \operatorname{logit} L_{js}}{2 \times 1.96}\right]^{-2}, \qquad c_s = \frac{\sum_j w_{js} x_{js} y_{js}}{\sum_j w_{js} x_{js}^2}.$$

The grid estimate is $\widehat{c}_{\mathrm{grid}} = \sum_s n_s c_s / \sum_s n_s$, with n_s the retained samples over the set's eight fields. Resampling the twenty pattern sets 20,000 times, with reference means $\bar{m}^1_s$ held fixed, gives $\widehat{c}_{\mathrm{grid}} = 0.1994$ and the 2.5th–97.5th percentile interval $[0.1908, 0.2095]$. Three alternative estimators give 0.1994–0.1997: weights $n_{js}\tilde{q}_{js}(1-\tilde{q}_{js})$, with $\tilde{q}_{js}$ the Wilson-interval center; all fields pooled before fitting; and a pooled binomial logistic fit $\operatorname{logit} q = \alpha + cx$ (intercept 0.000944).

Pattern-count and size comparisons. Section 5 compares settings with fewer runs, at a single field $\pm G$ and without a Hopfield reference. With $n_{+,s}(G)$ and $n_s(G)$ the positive and total retained counts of pattern set s at field G, let

$$S_s = n_{+,s}(+G) + n_s(-G) - n_{+,s}(-G), \qquad R_s = n_s(+G) + n_s(-G).$$

The pooled estimator, which assumes $q_{\mathrm{chain}}(G) + q_{\mathrm{chain}}(-G) = 1$ (checked only for $p = 3$, $N = 2048$; Table 4),

$$\widehat{c}_{\mathrm{scan}} = \operatorname{logit}\Big(\sum_s S_s \Big/ \sum_s R_s\Big) \Big/ \operatorname{logit} q_{\mathrm{eq}}^{\mathrm{CW}}(N),$$

uses the CW reference in place of the Hopfield one and does not fit a slope per pattern set. The $p = 3$, $N = 2048$ row of Table 1 comes from separate runs on twelve of the field grid's pattern sets ($s = 3$–14). One-pattern runs ($s = 0$; full protocol with the filtered start; $|G| = 0.0006125$) give c equal to 1.065, 1.126, 1.063, and 1.020 times $c_{\mathrm{CW}} = 0.227$, 0.171, 0.120, 0.090 at $N = 512$, 1024, 2048, 4096. Scaled by these ratios, the $p = 3$ prediction becomes 0.319 at $N = 1024$ and 0.153 at $N = 4096$; measured c remains 18% below and 4% above it (computed from unrounded values).

Table 5: Runs and seeds. Pattern set s is drawn with NumPy `default_rng`$(20260823 + 1009s)$. Each Hopfield run uses its own segment of a seeded PyTorch GPU generator. Bootstraps draw 20,000 pattern-set resamples and keep both field signs of a set together.

Experiment	Runs, steps, and seeds
Fig. 1a paths	Noise $\sqrt{2/N}$; 14 paths from $m = 0$ at $N = 1024$; 6 sweeps at each of 360 equally spaced β from 0.7 to 1.35; Euler step 0.02; NumPy seed 4
CW linear ramps	β piecewise constant at the midpoint of each interval $\Delta\beta = 0.005$, held for $0.005/\gamma$ expected sweeps; each interval has $0.25/\gamma$ (rounded) Richardson-extrapolated implicit Euler steps of about 0.02
Field grid	$s = 0$–19; 2000 initial runs per set and field; nine fields including $G = 0$; dynamics and bootstrap seed 20260825
Equilibrium reference	$s = 0$–19; 256 runs per set; dynamics seed 20260823
Pattern-count and size scans	$s = 3$–14; 8000 initial runs per set and field sign; dynamics seed $9000 + p$ (pattern count) or $4000 + N$ (size, $N \neq 2048$); bootstrap seed 2026 (pattern count) or 7 (size)
One-pattern runs	$s = 0$; 150,000 initial runs per field sign; $N = 512$–4096; dynamics seed $31337 + N$

Appendix C. Amplification and pattern correlations

This appendix derives the gain A_p of Eq. (6) and examines two weak points of the prediction: two patterns and pattern correlations.

Conditioning on the winning pattern. Eq. (6) follows from conditioning on the pattern that wins. Under the independent-Gaussian approximation, let $m^1 \sim \mathcal{N}(\delta, \sigma^2)$, where δ is the field-induced mean shift and the other overlaps have zero mean and variance σ^2. The density f of m^1 gives $f(r)/f(-r) = \exp(2r\delta/\sigma^2)$ at magnitude $r > 0$. Winning depends only on the magnitude, so it does not change these sign odds at fixed r:

$$\Pr(m^1 > 0 \mid |m^1| = r, \text{ pattern 1 wins}) = \frac{1}{1 + e^{-2r\delta/\sigma^2}} \simeq \frac{1}{2} + \frac{r\delta}{2\sigma^2}.$$

At $\delta = 0$, all patterns have the same distribution, so the winning magnitude is distributed as σM_p, where $M_p = \max_{1 \le \mu \le p} |Z_\mu|$ for independent standard Gaussians Z_μ. The magnitude distribution is even in δ, so its first-order change is zero. Averaging the sign probability over winning magnitudes then gives, to first order,

$$q_{\text{chain}} \simeq \frac{1}{2} + \frac{\delta}{2\sigma}\,\mathbb{E}M_p.$$

Expanding the logit around $1/2$ gives

$$\operatorname{logit} q_{\text{chain}} \simeq \frac{2\delta}{\sigma}\,\mathbb{E}M_p.$$

With one pattern, $M_1 = |Z|$. Dividing by this one-pattern response, with the same equilibrium-log-odds reference, gives Eq. (6). The calculation averages sign probabilities before taking log-odds. It gives sign odds given the winning pattern, not all pattern-pair frequencies.

Two patterns. For $p = 2$, the largest miss in Table 1, an exact decomposition gives a sharper estimate. With $y_i = \xi_i^1 x_i$, the spins on which the two patterns agree and those on which they disagree form two non-interacting CW blocks, of 980–1068 spins in our pattern sets. Pattern 1 wins exactly when the two blocks take the same sign, and their odds then multiply. Treating the blocks as independent and equal, each a CW model of about $N/2 = 1024$ spins, therefore doubles the logit 0.343384 of the one-pattern runs at $N = 1024$. Dividing by the exact CW $\operatorname{logit} q_{\mathrm{eq}} = 3.575$ at $N = 2048$ gives $c = 0.19212$, against the measured 0.18783 $[0.184, 0.191]$ and the Gaussian prediction 0.169. The block sizes and initialization remain unmatched.

Pattern correlations. Section 5 names pattern correlations as one possible cause of the size trend; this paragraph shows that the transition shifts they cause stay comparable to the critical window as N grows. For p patterns, $C_{\mu\nu} = N^{-1} \sum_i \xi_i^\mu \xi_i^\nu$ records their pairwise correlations. For continuous-time random-site dynamics at zero field, averaging the expected Gibbs redraw over sites, weighted by ξ_i^μ, gives

$$\dot{m}^\mu \simeq -m^\mu + \frac{1}{N} \sum_i \xi_i^\mu \tanh\left(\beta \sum_\nu \xi_i^\nu m^\nu\right) \simeq -m^\mu + \beta \sum_\nu C_{\mu\nu} m^\nu .$$

The first approximation neglects the self-exclusion correction of order p/N; the second uses $\tanh z \simeq z$ for small overlaps. The drift of $\boldsymbol{m} = (m^1, \ldots, m^p)$ is thus $(\beta C - I)\boldsymbol{m}$. Along an eigenvector of C with eigenvalue λ, an overlap grows at rate $\beta\lambda - 1$, beginning at $\beta = 1/\lambda$ rather than at $\beta = 1$ as for $C = I$. Each off-diagonal entry averages N independent random signs, so its standard deviation is $N^{-1/2}$. For fixed p, the eigenvalue shifts and the resulting transition shifts have this order, comparable to the critical window of Section 5. Separating these shifts from the selection range $\gamma T \sim \sqrt{\gamma}$ would require $N\gamma \gg 1$. This comparison does not identify the cause of the measured size trend.